

Student Academic Risk Prediction

Presentation Guide: How It Works

How to Run the Project

1. Double-click the 'run_project.bat' script in the project directory. This script will automatically install all necessary packages and start the dashboard.
2. The system will open a browser tab to 'http://localhost:8501' displaying the interactive app.

How to Present to the Teacher

Step 1: Introduction

- Explain the problem: Universities need a way to proactively identify students struggling academically.
- Explain your solution: A machine learning model that takes in student behaviors and predicts risk.

Step 2: Dashboard Demonstration

- Navigate to the 'Dashboard' page. Show the high-level statistics and Risk Distribution graph.
- Go to the 'Dataset Explorer' to show that you thoroughly understand the dataset features.

Step 3: Live Prediction

- Navigate to 'Predict Student Risk'.
- Enter hypothetical student data (e.g., very low attendance, low midterm score).
- Click 'Predict Risk' and show how the model correctly flags the student as 'At Risk'.

Step 4: Model Performance (The 'Data Science' part)

- Go to the 'Model Performance' page.
- Show the metric table and explain how you trained three models and selected the best one based on F1-Score.
- Highlight the Feature Importance graph, noting that 'Attendance' and 'Midterm Score' are heavy indicators.